Enrollment No.: _____

Darshan UNIVERSITY
योगः कर्मसु कौशलम्

Darshan Institute of Engineering & Technology
B.Tech. | Sem-5 | Winter-2024

Course Code	:	2301CS501	**Date**	:	17-10-2024
Course Name	:	Computer Network	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:
1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1 **(A)** Write a short note on LAN, MAN and WAN. **4**

(B) Compare OSI reference model and TCP/IP protocol suite. **3**

OR

Define following terms:
Computer Network, Processing Delay, Queuing Delay.

(C) Sketch the diagram of OSI reference model and discuss functionalities of all the layers. **7**

OR

Define Network Topology. List all types of topologies. Discuss the concepts of all network topologies.

Q.2 **(A)** Write the full form of DNS. List all types of DNS Components. Explain any 2 components. **4**

(B) What is the need of FTP? Discuss working of FTP. **3**

OR

Distinguish persistent and non-persistent http.

(C) Discuss the DORA process in DHCP. **7**

OR

List the protocols which are used in email. Explain mail access/receiving protocols with diagram.

Q.3 **(A)** Draw and discuss each field of TCP header. **4**

(B) Compare connection-oriented and connection less protocol. **3**

OR

What is the main reason to use sliding window protocol? Draw and discuss sliding window protocol.

(C) Explain the process of connection-establishment and connection release in terms of TCP. **7**

OR

Discuss the concepts of Multiplexing and demultiplexing in transport layer with appropriate diagram.

Q.4 **(A)** Draw IPV4 header format and explain the functionality of each field of IPV4 header. **4**

(B) Make a list of IP address class with its range. What are the default subnet mask of class A, B & C. Draw and explain network id and host id in class A, B & C. **3**

OR

Distinguish between IPV4 address and IPV6 address.

(C) Explain Routing Information Protocol with appropriate diagram. **7**

OR

Discuss Link state routing protocol with proper diagram.

Q.5 **(A)** Discuss the concept of variable size framing in terms of character oriented and Bit oriented with example. **4**

(B) Draw and discuss ethernet frame structure. **3**

OR

Write short note on random access collision sense protocol for collision detection and collision avoidance.

(C) A bit stream is transmitted 1101101 using the CRC method. The generator polynomial is X^4+X^2+1. What is the actual bit stream transmitted? **7**

OR

Calculate the checksum of given frame:
Frame1 - 11001100, Frame2 - 10101010, Frame3 - 11110000, Frame4 – 11000011.
Justify your answer whether data accepted or rejected at receiver side?

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